

Student Grading System using HTML, CSS and JavaScript

Experiment Report — Internal, External and Inline CSS/JavaScript

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Aim

To design a Student Grading System web page using HTML and CSS, and to demonstrate the use of external, internal, and inline CSS along with external and internal JavaScript for input validation, grade calculation, and DOM manipulation.

Software / Tools Required

- Visual Studio Code
- Google Chrome (or any modern web browser)
- HTML5
- CSS3
- JavaScript

Program Code

The page combines all three CSS types — **external** (style.css), **internal** (<style> block in the <head>), and **inline** (the style attribute on the "Calculate Grade" button) — along with **external JavaScript** (script.js, which validates the marks and computes the grade) and **internal JavaScript** (console messages logged when the page loads).

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Grade Calculator</title>

  <!-- External CSS -->
  <link rel="stylesheet" href="style.css">

  <!-- Internal CSS -->
  <style>
    h1{
      color:darkblue;
      text-align:center;
    }

    .container{
      width:350px;
      margin:auto;
      background:white;
      padding:20px;
      border-radius:10px;
      box-shadow:0px 0px 10px gray;
    }

    #result{
      margin-top:20px;
      font-size:20px;
```

```

        font-weight:bold;
        color:green;
    }
</style>
</head>

<body>

<div class="container">

<h1>Student Grading System</h1>

<label>Student Name</label><br>
<input type="text" id="name"><br><br>

<label>Enter Marks (0-100)</label><br>
<input type="number" id="marks"><br><br>

<!-- Inline CSS + Inline JavaScript -->
<button
style="background:blue;color:white;padding:8px 15px;"
onclick="calculateGrade()">
Calculate Grade
</button>

<button type="reset">Clear</button>

<div id="result"></div>

</div>

<!-- External JS -->
<script src="script.js"></script>

<!-- Internal JavaScript -->
<script>
console.log("Student Grade Page Loaded");
console.info("Enter marks between 0 and 100");
console.warn("Invalid marks are not accepted");
</script>

</body>
</html>

```

style.css

```

body{
    background:#d6ecff;
    font-family:Arial;
}

input{
    width:100%;
    padding:8px;
}

button{
    cursor:pointer;
    margin-top:10px;
}

```

script.js

```

function calculateGrade()
{
    var name = document.getElementById("name").value;
    var marks = parseInt(document.getElementById("marks").value);

```

```

if(name=="")
{
    alert("Enter Student Name");
    return;
}

if(isNaN(marks))
{
    alert("Enter Marks");
    return;
}

if(marks<0 || marks>100)
{
    alert("Marks should be between 0 and 100");
    return;
}

var grade;

if(marks>=90)
    grade="A+";
else if(marks>=80)
    grade="A";
else if(marks>=70)
    grade="B";
else if(marks>=60)
    grade="C";
else if(marks>=50)
    grade="D";
else
    grade="Fail";

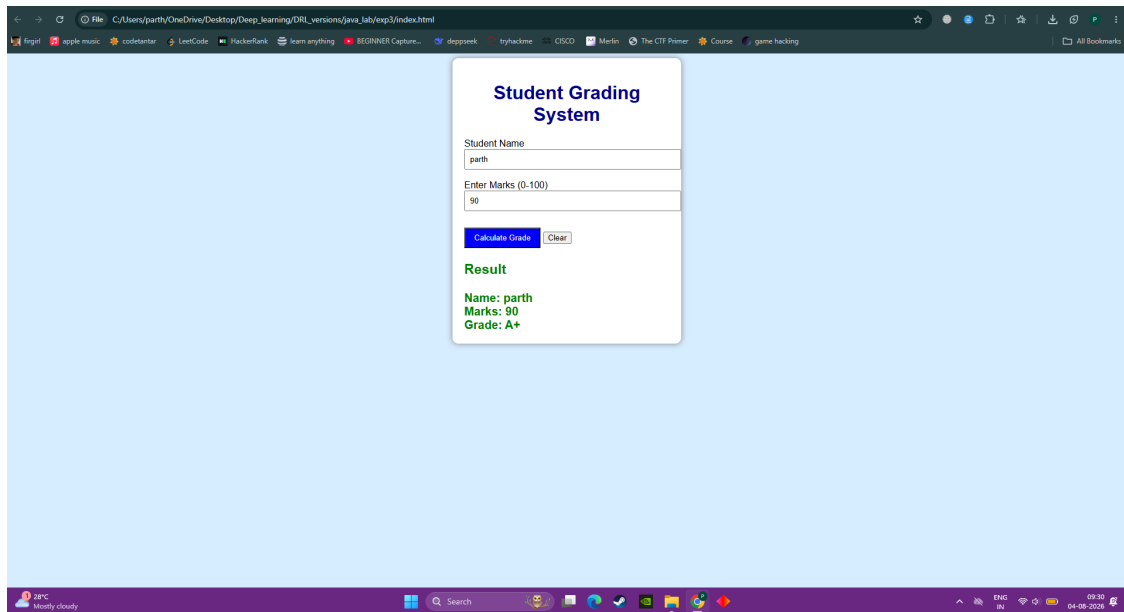
console.log("Student:", name);
console.log("Marks:", marks);
console.log("Grade:", grade);

document.getElementById("result").innerHTML =
"<h3>Result</h3>" +
"Name: " + name + "<br>" +
"Marks: " + marks + "<br>" +
"Grade: " + grade;
}

```

Output

The screenshot below shows the Student Grading System with a sample name and marks entered, displaying the computed grade.



Result / Conclusion

Result:

The Student Grading System was developed successfully using HTML, CSS, and JavaScript. The program validates the entered name and marks, and correctly computes and displays the grade (A+, A, B, C, D, or Fail) based on the marks entered. For example, a student named "parth" with 90 marks was correctly graded as A+.

Conclusion:

This experiment demonstrates the practical use of external, internal, and inline CSS together in a single page, and the combined use of external and internal JavaScript. It also reinforces concepts of input validation, conditional logic for grade computation, DOM manipulation using `getElementById`, and console-based debugging output.